8 Features of manufacturing industries

8.3 Modern manufacturing methods and systems when designing for manufacture for small, medium and large-scale production

Topic	What students need to learn:
8 Features of manufacturing industries	8.3 – Characteristics, processes, application, advantages and disadvantages and the importance of considering accuracy of production and efficiency of modern manufacturing methods and systems when designing for manufacture for small, medium and large scale production: *a) production scheduling and production logistics* *b) robotics in production – robots on fully-automated production and assembly lines/cells*

Topic	What students need to learn:
8 Features of manufacturing industries	*c) materials handling systems – automated storage and retrieval systems (ASRS), automatic guided vehicles (AGVs)* *d) flexible manufacturing systems (FMS), modular/cell production systems* *e) lean manufacturing using just-in-time (JIT) systems* *f) standardised parts, bought-in components* *g) quick response manufacturing (QRM)* *h) data integration – product data management (PDM), enterprise resource planning (ERP) systems* *i) concurrent manufacturing.*

i) Concurrent Manufacturing

Concurrent Manufacturing

- In order to remain competitive and cope with increasing market pressure from mounting customer demands, manufacturers need to get to market first with products that customers want.
- **Concurrent manufacturing** is about all of the key people who work at each stage of the design and manufacturing process working together to make sure that changes to one part of the process will not require changes to be made at another.
- Designs become 'right first time'.
- This reduces product development times and enables the earlier release of new products.
- **Concurrent manufacturing systems** aim to eliminate the need for design changes and overcome production problems and product introduction delays.

Concurrent Manufacturing

- Delays and problems add significant costs to a product which in turn makes it less competitive.
- **Concurrent manufacturing systems** bring together members from a wide range of disciplines – such as design, manufacturing, project management, technical support, marketing and other specialist areas to form a multidisciplinary team.
- By working together, differences are more easily dealt with early in design and production, making the design process quicker and more efficient.
- Effectiveness of the team is important.
- Excellent communication is needed to be effective.
- Computer-based systems enable efficient communication between individual team members and integrated project teams for product development.

Concurrent Manufacturing

- For manufacturing companies operating at a national or international level, a computer network is essential to support data transfer between team members.
- *Concurrent manufacturing systems, effective management and teamwork ensure the development of a high-quality product at the right price in the shortest possible time.*

Summary - Concurrent Manufacturing

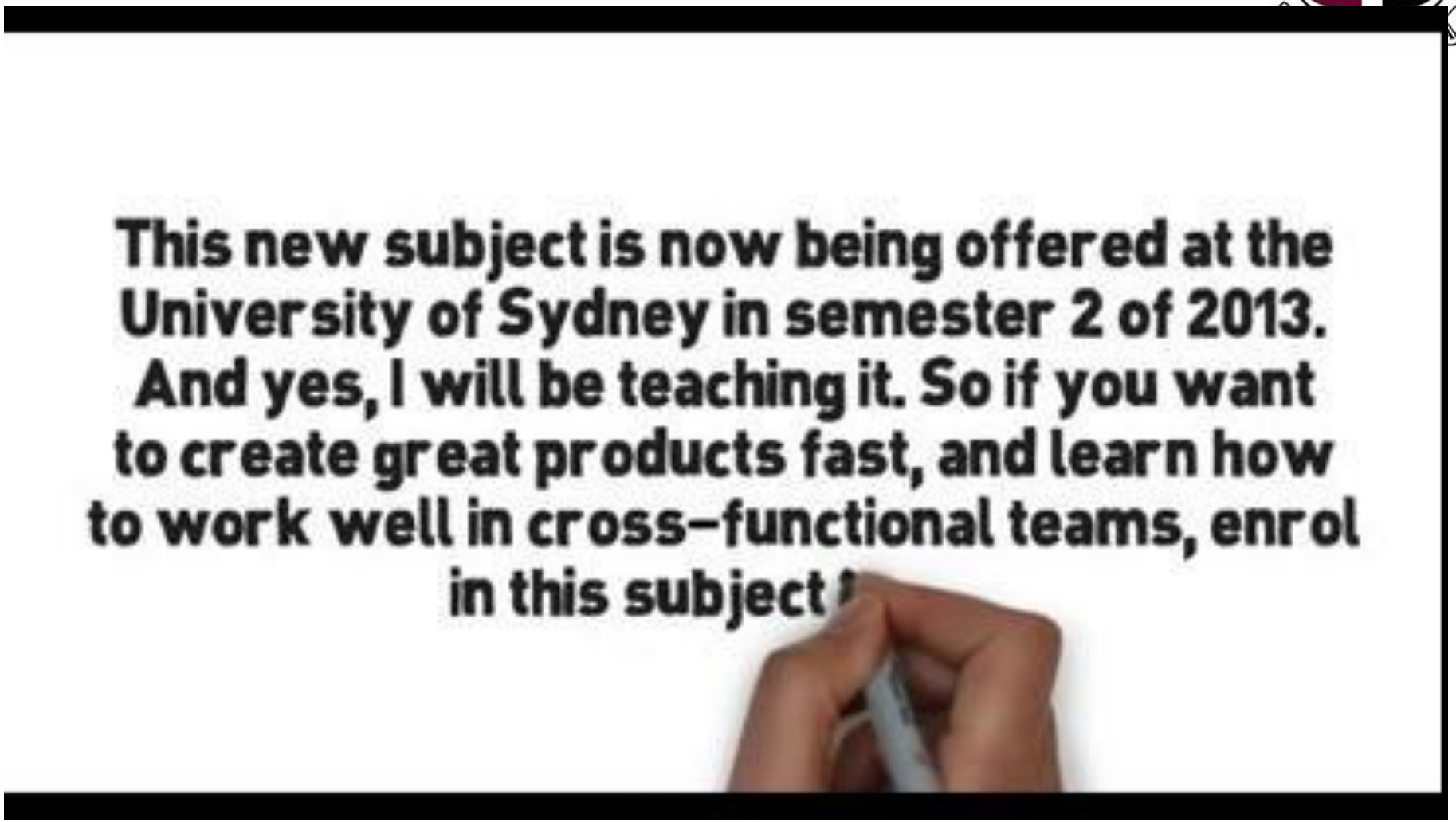
- **Concurrent Manufacturing** is a method of manufacturing which utilises the parallel processing of tasks to bring products to the market more quickly.
- The traditional approach to design and manufacture was a ***linear system*** in which each department would conduct their element of the process and then pass it on to the next one.
- This method often resulted in delays, low quality, poor communication between departments and less competitive products.
- Concurrent manufacturing shortens the time it takes to bring high quality products to the market by the systematic integration of all the elements of design and manufacture so that they can be carried out simultaneously wherever possible.

Comparison of *Linear* and *Concurrent* Manufacturing Systems

Traditional Linear System (Planning)	Timescale									
	1	2	3	4	5	6	7	8	9	10
Project initiation										
Design										
Materials Sourcing										
Manufacture										
Delivery										

Concurrent System (Planning)	Timescale									
	1	2	3	4	5	6	7	8	9	10
Project initiation										
Design										
Materials Sourcing										
Manufacture										
Delivery										

Video: Implementing Concurrent Projects



Key features of Concurrent Manufacturing

- Uses multi-disciplinary teams to produce high quality designs that are ‘right first time’.
- Computer networks and related technology are important to facilitate communication.
- Design issues can be resolved early in the process.
- Facilitates the agreement of compromises where necessary.
- Manufacturing can start earlier, even before final details of some elements are decided.
- **Quality Function Deployment (QFD)**, a quality assurance system, can be used in association with concurrent manufacturing to ensure that consumer requirements are met.
- QFD ‘House of Quality’ planning systems can be incorporated to assure high quality – using a ***‘matrix’***.

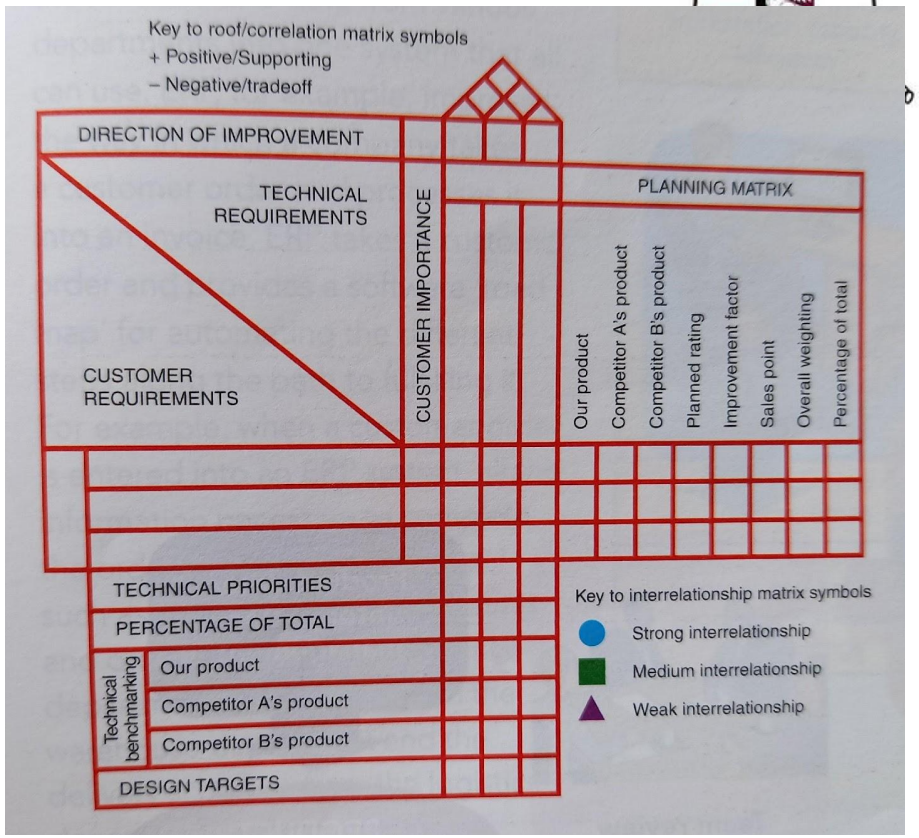
Concurrent Manufacturing

Definition:

- **Quality Function Deployment (QFD)** is a method that incorporates customer satisfaction into the development of a product before it is manufactured.
- The main features of QFD are its focus on customer requirements, the use of multi-disciplinary teamwork and a comprehensive *'House of Quality' matrix*.
- **QFD 'House of Quality' matrix** is used by the team to translate customer requirements into a number of targets or specifications to be met by the new product. Advantages:
    - *Reduce time to market*
    - *Reduction in design modifications*
    - *Decreased design and manufacture costs*
    - *Improved product quality*
    - *Enhanced customer satisfaction*

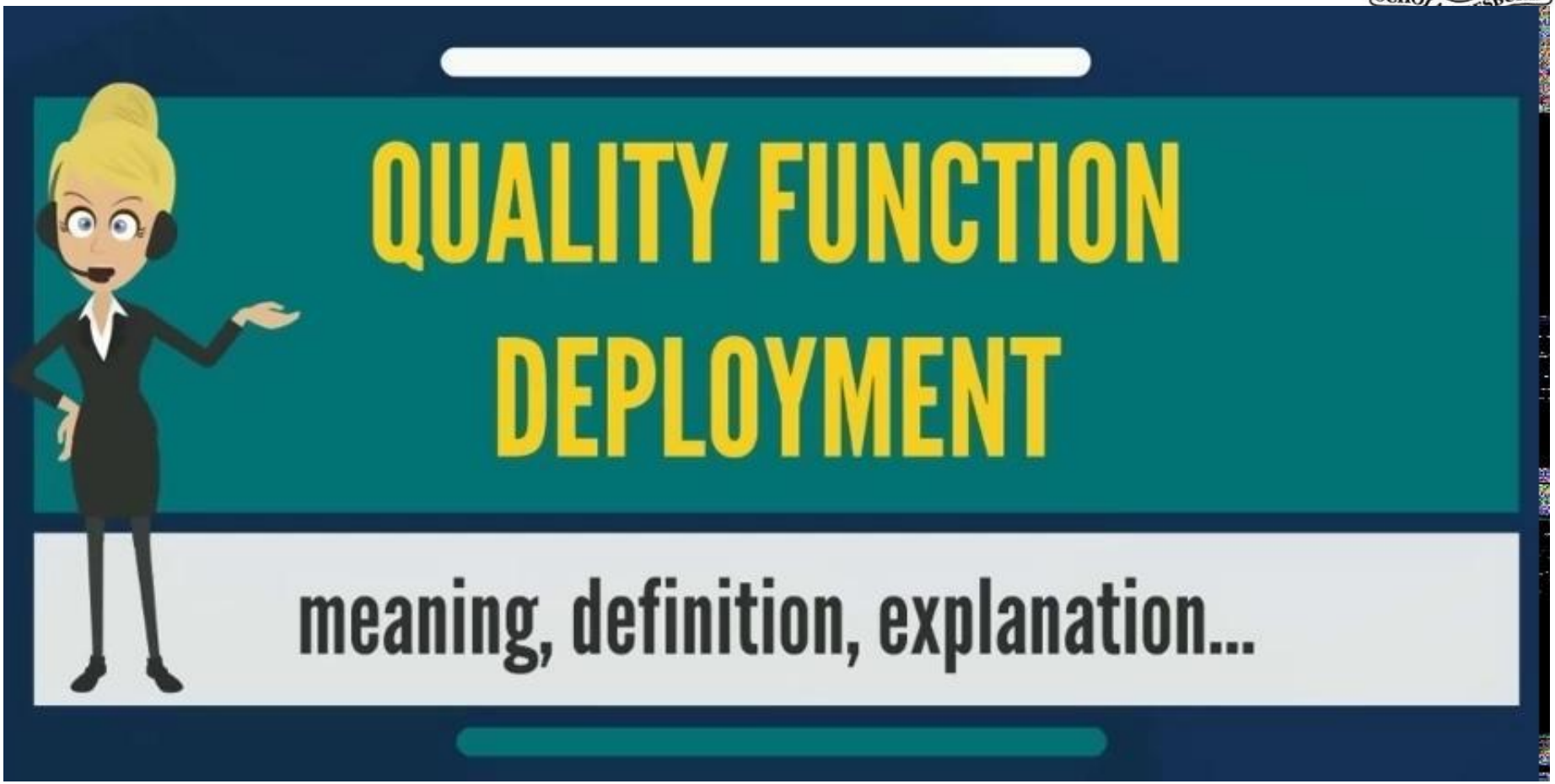
The six major components of a ‘House of Quality’ matrix

Component		Content
1	Customer requirements	A structured list of requirements taken from customer statements in a market survey
2	Technical requirements	A technical specification is produced
3	Planning matrix	A plan of what the product needs to meet customer and technical demands
4	Interrelationship matrix	A grid showing how technical requirements dovetail with customer needs
5	Technical matrix	An analysis of how the various technical requirements support or impede each other. E.g. a high-quality finish such as 300 DPI and the need for low-cost paper would demand a compromise or possibly a change to the required printing process
6	Technical priorities, benchmarks and targets	Quality targets against each technical specification point can be measured. E.g. the colour density on a print run

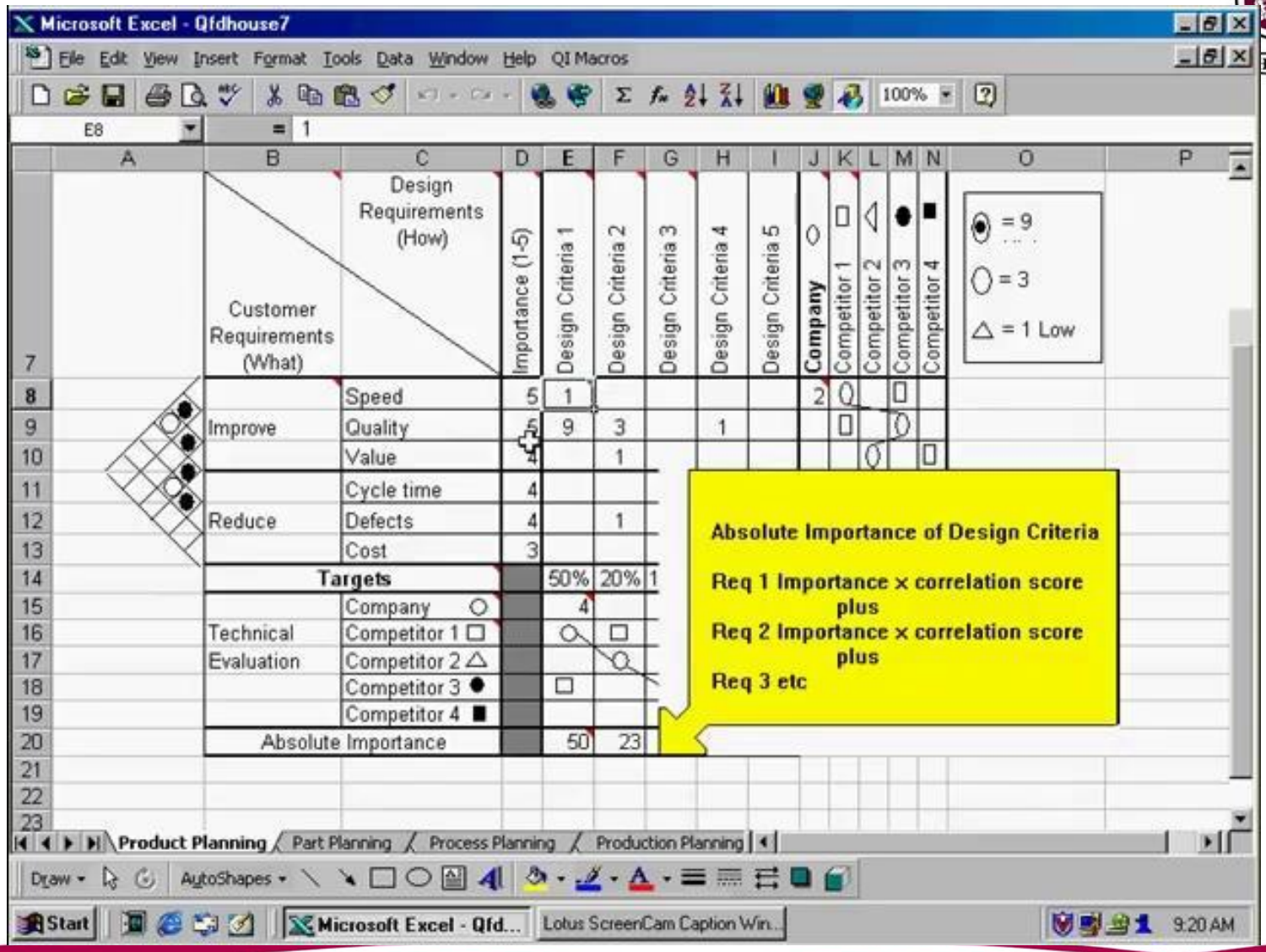


Example of a ***‘House of Quality’ matrix*** for assuring the quality of a design

Video: Quality Function Deployment (QFD)



Video: QFD House-of-Quality Matrix



Exam Tip:

A recurring theme this chapter is the ‘flexibility’ and ‘fast response’ that is now possible through the implementation of modern manufacturing technology.

You need to understand the link between what you have learned here and strategies such as ***‘mass customisation’*** seen in chapter 8.1, which blur the traditional distinction between scales of production.

Exercise: if you want to see what ***‘mass customisation’*** looks like click on the link below and ***‘configure’*** you own Mini. Most car manufacturers are able to mass produce cars to buyers exact specifications.
https://configure.mini.co.uk/en_GB/model-ranges/F56

Now Test Yourself:

1. Explain briefly how ***concurrent manufacturing*** can bring a product to market more quickly than by using a conventional ***linear system***.